PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review
AUTHORS	Ding, Hanyue; Huang, Junjie; Deng, Yunyang; Tin, Pamela; Wong, Martin Chi-Sang; Yeoh, EK

VERSION 1 – REVIEW

REVIEWER	Lowe, JM The University of Newcastle, University of Newcastle
REVIEW RETURNED	25-Oct-2021

GENERAL COMMENTS	Thank you for identifying an important topic for review. However i have some concerns that your search strategy may have been inadequte and has missed numerous national NCD surveys following the WHO STEPS studies stage 111 of which indudes blood for l lipids and glucose measurement: for example Tripathy JP et al Daibetologiy and metabolic Syndrome 2017;9:8 and Msyamboza KP et al PLOS One pu May 23 2011 . If these have been identified and excluded for some reason this should be made clear. I doubt if inclusion would make much difference to your conclusions but it would strengthen them. I was also unclear as to why the study by Sargeant (ref 61) was referred to but not included, but it is possible that another study using the same data was included. A minor suggestion in Fig 1 you could shortern the text to "38 studies in 33 articles included".

REVIEWER	Lindblad, Ulf Sahlgrenska Academy at Gothenburg University, Public Health and Com. Med./Primary Care
REVIEW RETURNED	16-Dec-2021

GENERAL COMMENTS	Cardiovascular disorders and their complications are dominating global causes of morbidity and mortality. Accordingly, there is an interest to screen populations for identification of subjects of interest for preventive interventions. With that background the authors made a systematic review of relevant papers to learn more of characteristics of participants in diabetes and/or lipid screening to better direct future surveillance and intervention. Identification of relevant publications was based on a structured search of papers published in Pub Med or Medline between 1st Jan 2000 and 1st April 2020 for specified criteria identifying screening for diabetes or lipids. The authors found 5,723 specific citations meeting their search criteria and after further review 38

	studies in 33 articles were included in the present systematic review. Data extraction focused on socioeconomic and demographic characteristics. Generally, more women than men participated in specific diabetes screening (11 studies). These participants tended to have a higher BMI than their respective national mean as reported by WHO. There was a lower participation among those with low education and those who were socially deprived, current smokers or alcohol drinkers. Only 4 studies reported specific lipid screening and among these both men and women had a lower BMI than their national mean. In all, 18 studies reported data from combined diabetes and lipid screening. In these studies, there was a higher participation of females, and while they generally had a higher BMI than their national means, the opposite was found in men. Reasons for non-participation varied, but lack of time and no interest were most frequently reported. Subjects with a lower socio-economic status were less likely to respond to screening in developed areas. While there were several differences in details between specific studies as discussed, the general pattern still resembles what is previously known from epidemiological surveys within this field. Still, the major contribution comes from the extensive current literature search, representing a wide range of different cultural, socio-economic and demographic backgrounds. Comments 1. While s major part of cardiovascular morbidity is commonly attributed to hypertension, this was not included in the present review. Thus, this should be addressed and discussed accordingly: - The rational for excluding hypertension. - The potential limitations to the implementation of the study findings. 2. The descriptive findings are presented accordingly, however, I recommend a more specific presentation of what is new facts compared to previous knowledge. 3. It want to see the outcome presented also by cultural and ethnic background. Ulf Lindblad

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. JM Lowe, The University of Newcastle, Sunnybrook Health Sciences Centre
Comments to the Author:
Thank you for identifying an important topic for review. However i have some concerns that your search strategy may have been inadequte and has missed numerous national NCD surveys following

the WHO STEPS studies stage 111 of which indudes blood for l lipids and glucose measurement: for example Tripathy JP et al Daibetologiy and metabolic Syndrome
2017;9:8 PubMed and Msyamboza KP et al PLOS One pu May 23 2011 . If these have been identified and excluded for some reason this should be made clear. I doubt if inclusion would make much difference to your conclusions but it would strengthen them. I was also unclear as to why the study by Sargeant (ref 61) was referred to but not included, but it is possible that another study using the same data was included.

Response: Thank you for the suggestion. We have optimized the inclusion and exclusion criteria. Since this review only included the one-step tests for measuring lipid and/or diabetesscreening, the studies with stepwise measurements may show different characteristics of participants with our study. We also clarified the description of the study by Sargeant (ref 61) in the discussion part.

Revision:
Line 168
3) adopted one-step finger stick or blood glucose (including fasting plasma glucose (FPG), oral glucose tolerance tests (OGTT), glycated hemoglobin (HbA1c), random plasma glucose (RBG)) or blood lipid tests as screening tools;
Line 176-177
4) studies involved the measurements with many steps/stages; and 5) studies performed before 2000.
Line 412
Sargeant and colleagues conducted a stepwise program to screen diabetes, and they reported male gender was significantly associated with lower uptake of RBG testing, which is consistent with our findings on diabetes screening.

A minor suggestion in Fig 1 you could shortern the text to "38 studies in 33 articles included".

Response: Thank you for the comment. We have revised this figure.

Revision: Supplementary figure 1

Reviewer: 2
Prof. Ulf Lindblad, Sahlgrenska Academy at Gothenburg University
Comments to the Author:
Cardiovascular disorders and their complications are dominating global causes of morbidity and mortality. Accordingly, there is an interest to screen populations for identification of subjects of interest for preventive interventions. With that background the authors made a systematic review of relevant papers to learn more of characteristics of participants in diabetes and/or lipid screening to better direct future surveillance and intervention.

Identification of relevant publications was based on a structured search of papers published in Pub Med or Medline between 1st Jan 2000 and 1st April 2020 for specified criteria identifying screening for diabetes or lipids. The authors found 5,723 specific citations meeting their search criteria and after further review 38 studies in 33 articles were included in the present systematic review.

Data extraction focused on socioeconomic and demographic characteristics. Generally, more women than men participated in specific diabetes screening (11 studies). These participants tended to have a higher BMI than their respective national mean as reported by WHO. There was a lower participation among those with low education and those who were socially deprived, current smokers or alcohol drinkers.

Only 4 studies reported specific lipid screening and among these both men and women had a lower BMI than their national mean. In all, 18 studies reported data from combined diabetes and lipid screening. In these studies, there was a higher participation of females, and while they generally had a higher BMI than their national means, the opposite was found in men.

Reasons for non-participation varied, but lack of time and no interest were most frequently reported. Subjects with a lower socio-economic status were less likely to respond to screening in developed areas.

While there were several differences in details between specific studies as discussed, the general pattern still resembles what is previously known from epidemiological surveys within this field. Still, the major contribution comes from the extensive current literature search, representing a wide range of different cultural, socio-economic and demographic backgrounds.

Comments

1. While s major part of cardiovascular morbidity is commonly attributed to hypertension, this was not included in the present review. Thus, this should be addressed and discussed accordingly:
- The rational for excluding hypertension.

Response: Thank you for the comment. We have added this part in the discussion.

Revision: Line 455-459
Hypertension is also a risk factor to contribute the CVD, but we did not include it as a target measurement. Because the blood tests had more obstacles than the blood pressure tests, for example, the fear of needles, fear of pain, and cost of test strips and needles.[68] Thus in this study, we focused on the blood glucose tests and lipid profiles to explore the characteristics of participants.

- The potential limitations to the implementation of the study findings.

Response: Thank you for the comment. The limitations were shown in Line 464-475.

2. The descriptive findings are presented accordingly, however, I recommend a more specific presentation of what is new facts compared to previous knowledge.

Response: Thank you for the comment. To our knowledge, this is the first systematic review that summarized the characteristic of participants who take up screening tests for diabetes and lipid disorders. We summarized our results (Line 402-410), compared our results with previous studies (Line 412-453) and emphasized the findings in the discussion part. (Line 477-487)

3. It want to see the outcome presented also by cultural and ethnic background.

Response: Thank you for the comment. We have added some ethnic backgrounds in table 1.

VERSION 2 – REVIEW

REVIEWER	Lowe, JM The University of Newcastle, University of Newcastle
REVIEW RETURNED	27-Jan-2022

GENERAL COMMENTS	Thank you for your revisions. I have some additional comments, mostly minor. In the Abstract I suggest you delete "who" in line 47. In the strengths and Limitations section I suggest you add "one-step" before "diabetes" in line 60 In the main article: line 151 suggest change "many" to "multiple" Line 180. I strongly disagree with the statement about public involvement-it is exactly this attitude that restricts it. At this stage I think you should at least include plans for dissemination to the public for example through national, regional, local diabetes and heart disease support groups, otherwise the issues concerning groups underrepresented will never be resolved. Line 247 I think you have misunderstood the Table in the article by Zafar. "Matric" is the local term for the secondary school leaving certificate in Pakistan taken at the end of year 10 and I doubt "conventional " is the same as a postgraduate degree as pakistan has high rates of illiteracy and the mean number of years of education is just over 5 years. I suspect it refers to people educated in religious schools or through apprenticeships only. Please revise this section If you are going to comment on the low smoking rates in Studies in Nigeria and Pakistan it should be placed in context- national smoking rates in Nigeria are under 5% and in Pakistan around 20%. Similarly the very low rate of alcohol use in Nigeria is very unusual as problem alcohol use is reported to be around 50% or more, suggesting the population is unrepresentative (or not reporting their alcohol use). Line 431 i suggest a comma "," after "measurement" and lowercase "b" in "because"

REVIEWER	Lindblad, Ulf Sahlgrenska Academy at Gothenburg University, Public Health and Com. Med./Primary Care
REVIEW RETURNED	02-Feb-2022

GENERAL COMMENTS	I still don not think that the exclusion of hypertension is sufficiently discussed, however, I leave to the editors to decide on this. I they approve the revision I would be OK with that.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1

Dr. JM Lowe, The University of Newcastle, Sunnybrook Health Sciences Centre

Comments to the Author:

Thank you for your revisions. I have some additional comments, mostly minor.

Response: Thank you for your comments. We have revised these points according to your comments.

In the Abstract I suggest you delete "who" in line 47.

Response: Thank you for the comment. We have deleted it.

In the strengths and Limitations section I suggest you add "one-step" before "diabetes" in line 60

Response: Thank you for the comment. We have added it.

Revision: Line 60-61

To our knowledge, this is the first systematic review that presented the characteristics of screening participants in one-step diabetes and lipid screening programs.

In the main article:

line 151 suggest change "many" to "multiple"

Response: Thank you for the comment. We have revised it.

Revision: Line 152

4) studies involved the measurements with multiple steps/stages.

Line 180. I strongly disagree with the statement about public involvement-it is exactly this attitude that restricts it. At this stage I think you should at least include plans for dissemination to the public for example through national, regional, local diabetes and heart disease support groups, otherwise the issues concerning groups underrepresented will never be resolved.

Response: Thank you for the suggestion. As this is a systematic review, we just summarised all information in this study, but we plan to involve the public in our further study based on the results of this review.

Line 247 I think you have misunderstood the Table in the article by Zafar. "Matric" is the local term for the secondary school leaving certificate in Pakistan taken at the end of year 10 and I doubt "conventional " is the same as a postgraduate degree as pakistan has high rates of illiteracy and the mean number of years of education is just over 5 years. I suspect it refers to people educated in religious schools or through apprenticeships only. Please revise this section

Response: Thank you for the suggestion. We have revised Table 2 and the manuscript.

Revision: Table 2 and Line 247-250

Zafar et al showed more than half of participants had matriculation level or above (same as the 10th grade) or above (52.5%) and only 4.2% acquired conventional level (same as the postgraduate degree).

If you are going to comment on the low smoking rates in Studies in Nigeria and Pakistan it should be placed in context- national smoking rates in Nigeria are under 5% and in Pakistan around 20%. Similarly the very low rate of alcohol use in Nigeria is very unusual as problem alcohol use is reported to be around 50% or more, suggesting the population is unrepresentative (or not reporting their alcohol use).

Response: Thank you for the suggestion. We summarised the characteristic of participants who take up screening tests for diabetes and lipid disorders, which may not show the same characteristics as the general population. For smokers, we have discussed the lower uptake of tests, inferred the reason, and proposed interventions in Line 394-401.

Line 431 i suggest a comma "," after "measurement" and lowercase "b" in "because"

Response: Thank you for the suggestion. We have revised it.

Revision:Line 432.

Hypertension is also a risk factor to contribute the CVD, but we did not include it as a target measurement, because the blood tests had more obstacles than the blood pressure tests.

Reviewer: 2

Prof. Ulf Lindblad, Sahlgrenska Academy at Gothenburg University

Comments to the Author:

I still don not think that the exclusion of hypertension is sufficiently discussed, however, I leave to the editors to decide on this. I they approve the revision I would be OK with that.

Response: Thank you for the comment. We have discussed the exclusion of hypertension in Line 431-435. As the objective of this study is to explore the characteristics of participants who take up screening tests for diabetes and lipid disorders, we may consider the hypertension part in the further study.